

Conditional Statements: Answers (1-50)

Q1. What is a conditional statement in programming?

Answer: A conditional statement is used to make decisions based on whether a condition is true or false.

Q2. What does an if statement do?

Answer: An if statement executes a block of code when a specified condition is true.

Q3. What is the purpose of an else block?

Answer: An else block executes code when the if condition is false.

Q4. When is an else block executed?

Answer: It is executed when the condition in the if statement evaluates to false.

Q5. What is a Boolean expression?

Answer: A Boolean expression is an expression that returns either true or false.

Q6. Write the syntax of a simple if statement.

Answer: `if(condition) { // code }`

Q7. What operator is commonly used to test equality?

Answer: The equality operator (`==`) is used.

Q8. What does the modulus (%) operator return?

Answer: It returns the remainder after division.

Q9. How can you check if a number is even?

Answer: `number % 2 == 0`

Q10. How can you check if a number is odd?

Answer: `number % 2 != 0`

Q11. What is the purpose of else-if?

Answer: It allows testing multiple conditions one after another.

Q12. What is a nested if statement?

Answer: A nested if statement is an if statement placed inside another if statement.

Q13. Can an if statement exist without an else?

Answer: Yes, an if statement can exist without an else block.

Q14. What is the output when a false condition is tested in an if statement without else?

Answer: No code inside the if block is executed.

Q15. What is the role of braces {} in C/C++ conditionals?

Answer: Braces define the block of code that belongs to the conditional statement.

Q16. What does > mean?

Answer: Greater than.

Q17. What does < mean?

Answer: Less than.

Q18. What does >= mean?

Answer: Greater than or equal to.

Q19. What does <= mean?

Answer: Less than or equal to.

Q20. What does != mean?

Answer: Not equal to.

Q21. What is a switch statement?

Answer: A switch statement selects and executes one block of code from multiple choices.

Q22. What is a case label in a switch statement?

Answer: A case label represents a specific value to be matched in the switch statement.

Q23. Why is break used in switch?

Answer: Break stops execution after a matching case is executed.

Q24. What happens if break is omitted?

Answer: Execution continues to the next cases. This is called fall-through.

Q25. What is the default case in switch?

Answer: The default case executes when no case matches the switch expression.

Q26. Can switch compare strings in C++?

Answer: No, traditional C++ switch statements cannot directly compare strings.

Q27. Which statement is better for many constant choices: if-else or switch?

Answer: A switch statement is generally better.

Q28. How do you test whether a value is positive?

Answer: `value > 0`

Q29. How do you test whether a value is negative?

Answer: `value < 0`

Q30. How do you test whether a value is zero?

Answer: `value == 0`

Q31. Write a condition to check if age is at least 18.

Answer: `age >= 18`

Q32. Write a condition to check if marks are above 50.

Answer: marks > 50

Q33. What is decision making in programming?

Answer: Decision making is the process of choosing actions based on conditions.

Q34. Can nested if statements have multiple levels?

Answer: Yes, nested if statements can have multiple levels.

Q35. What is logical AND (&)?

Answer: It returns true only if both conditions are true.

Q36. What is logical OR (||)?

Answer: It returns true if at least one condition is true.

Q37. What is logical NOT (!)?

Answer: It reverses the Boolean value of a condition.

Q38. How do you check if a year is divisible by 4?

Answer: `year % 4 == 0`

Q39. What is a leap year?

Answer: A leap year has 366 days instead of 365 days.

Q40. What is the first condition for a leap year?

Answer: The year must be divisible by 4.

Q41. How do you find the largest of two numbers?

Answer: Compare them using an if-else statement.

Q42. How do you find the largest of three numbers?

Answer: Use if-else-if conditions to compare all three numbers.

Q43. What is an if-else-if ladder?

Answer: It is a series of conditions checked one after another until one becomes true.

Q44. What is the difference between if and switch?

Answer: If can evaluate complex conditions, while switch compares a single expression against constant values.

Q45. Can switch use ranges directly?

Answer: No, switch cannot use ranges directly.

Q46. What is fall-through in switch?

Answer: Fall-through occurs when execution continues to the next case because break is missing.

Q47. What data type is usually used for conditions?

Answer: The Boolean (bool) data type.

Q48. Why are conditional statements important?

Answer: They allow programs to make decisions and perform different actions.

Q49. Give one real-life example of a conditional statement.

Answer: If the traffic light is green, move forward; otherwise, stop.

Q50. Name four conditional structures covered.

Answer: If Statement, If-Else Statement, Else-If Ladder, Switch Statement.